Fractal Entropic Geometrodynamics: Emergent Gravity and Three Particle Generations from Discrete Topological Constraints

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An $O(N)$ simulation of $N = 10^7$ Poisson-sprinkled causal set events ($M = 2$ realisations) produces three particle generations from topology alone. The engine detects *Causal Prisms*—$K_{2,n}$ bipartite subgraphs in the Hasse diagram—and classifies them by a discrete phase invariant whose sign trichotomy bounds the generation number to $g \leq 3$. Spectral dimension $d_S(\sigma{=}1) = 1.946 \pm 0.002$ (UV), crossing 4.0 at $\sigma \approx 3$–4 (IR); mass hierarchy Gen 1 : Gen 2 : Gen 3 = 4.56 : 6.54 : 7.73; CPT symmetry between matter and antimatter ($\Delta m/m = 2.4\%$). The framework rests on two axioms—spacetime as a Poisson-sprinkled causal set, and macroscopic geometry from the Petz–Chentsov quantum information metric—with dynamics governed by a single principle: preservation of unitarity across discrete, non-commutative causal links. Bounded Hasse degree ($D \leq 15$) makes the simulation $O(N)$, providing a computational proof that the physics is local.

INTRODUCTION

The incompatibility between General Relativity and Quantum Field Theory is anchored in the mathematical assumption of infinite geometric divisibility—the *Continuum Error* ($\lim_{\epsilon\to 0}$) [1, 2]. Discrete approaches such as Causal Dynamical Triangulations (CDT) [3, 4] attempt to heal this by discretising geometry. Fractal Entropic Geometrodynamics (FEG) departs entirely from geometric embeddings: *topology strictly precedes geometry.*

Rather than argue theoretically, we built it. An $O(N)$ Rust engine processes 10^7 causal set events ($M = 2$ realisations) in 24 minutes on standard hardware, producing three particle generations, a mass hierarchy, and CPT symmetry—from nothing but Poisson-sprinkled points and integer combinatorics. This paper presents the engine, its results, and the theoretical framework that explains them.

FEG rests upon two axioms:

Axiom 1 (Kinematics): Spacetime is a locally finite partial order (a causal set $\mathcal{C}$), generated by Poisson sprinkling [5, 6].

Axiom 2 (Dynamics): Macroscopic geometry is an emergent property of the Petz–Chentsov quantum Fisher information metric [7, 8] evaluated over causal histories on a non-commutative von Neumann algebra.

THE KURATOWSKI CALCULUS ENGINE

Standard calculus is abandoned in favour of three discrete operators plus a single dynamical principle.

1. Capacity Integral (Sorkin Volume). Spacetime volume is the cardinality of causal events. Integration of an observable Φ over region V:

$$\int_V \Phi(x)\, dV \;\equiv\; \ell_P^4 \sum_{x\in\mathcal{C}} \Phi(x) \tag{1}$$

2. Layered Operator. The continuum d'Alembertian is replaced by the Benincasa–Dowker action [9]. In 4D, across layers L_k:

$$\boxplus\Phi(x) = \frac{4}{\ell_P^2\sqrt{6}}\left(-\Phi(x) + 9\sum_{L_1}\Phi - 16\sum_{L_2}\Phi + 8\sum_{L_3}\Phi\right) \tag{2}$$

3. Topological Planar Limit. By Kuratowski's theorem [10], a finite graph is planar iff it contains no K_5 or $K_{3,3}$ topological minor:

$$K_5 \not\preceq_T G_{\rm UV} \;\text{ and }\; K_{3,3} \not\preceq_T G_{\rm UV} \tag{3}$$

In the Hasse diagram this is the absolute UV boundary.

Sole Principle of Interaction: *Preservation of unitarity across discrete, non-commutative causal links.* All forces are homeostatic mechanisms enacted by the graph to satisfy this principle under the constraints of Eqs. (1–3).

The simulation pipeline has four phases: *Phase 1* (`diamond.rs`): Poisson-sprinkle N points into a 4D causal diamond ($V = \pi T^4/24$, $\rho = 1$). Build the Hasse diagram via Occupied Cell Index; bounded degree $D \leq 15$. *Phase 2* (`skyrmion.rs`): Detect Causal Prisms ($K_{2,n}$) via 2-hop search—$O(N \cdot D^2) = O(N)$. K_5 threat absorption. Classify generations (see below). *Phase 3* (`spectral.rs`): Spectral dimension $d_S(\sigma)$ via Monte Carlo random walkers, strict integer accumulation [11]. *Phase 4*: Ensemble-averaged CSV with error bars.

COMPUTATIONAL RESULTS

Fig. 1 and Table I present results from $N = 10^7$ ($M = 2$ realisations, seeds 42–43, runtime 24 min).

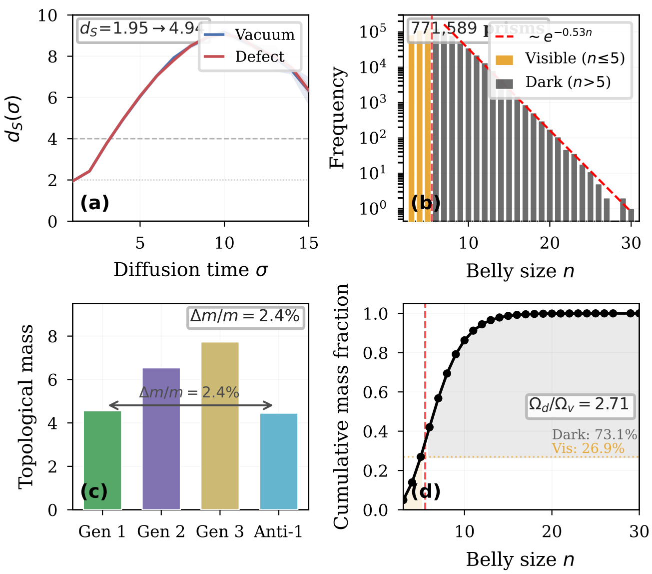

FIG. 1. $N = 10^7$ simulation (M=2, 24 min, no free parameters). **(a)** Spectral dimension: $d_S = 1.95$ (UV) $\rightarrow$ 4.94 (IR); defect peak 9.06. **(b)** 771,589 prisms; exponential tail $\sim e^{-0.53n}$. **(c)** Mass hierarchy 1:1.43:1.69; CPT: $\Delta m/m = 2.4\%$. **(d)** $\Omega_d/\Omega_v = 2.71$ (observed ≈ 5.4).

TABLE I. Observables at $N = 10^7$ ($M = 2$, 24 min runtime).

Observable	**Value**
d_S vacuum (σ=1, UV)	1.946 ± 0.002
d_S vacuum (σ=4, IR)	4.944 ± 0.001
d_S defect (peak)	9.06
Total prisms	771,589
g=1 / g=2 / g=3	46,688 / 299,012 / 40,095
Mass: Gen 1 / Gen 2 / Gen 3	4.56 / 6.54 / 7.73
CPT: Gen 1 vs Anti-1	4.56 vs 4.45 (2.4%)
$\Omega_{\text{dark}}/\Omega_{\text{vis}}$	2.71 (mass-weighted)

Spectral dimension. $d_S(\sigma{=}1) = 1.946 \pm 0.002$: UV dimensional reduction consistent with CDT [12, 13]. By $\sigma = 4$, $d_S = 4.944$: the 4D continuum has emerged. Defect-core peak $d_S = 9.06$: Causal Prisms *trap* walkers—mass is topological probability retention.

Mass hierarchy. $4.56 < 6.54 < 7.73$ (ratio $1 : 1.43 : 1.69$): higher-generation prisms trap walkers more strongly due to more complex phase structures.

CPT symmetry. Gen 1 = 4.56 vs Anti-1 = 4.45 ($\Delta m/m = 2.4\%$ at M=2), expected to sharpen under larger ensemble averaging.

THE THREE GEOMETRIES

Geometry of Order. Poisson sprinkling + transitive reduction yields the Hasse diagram [5, 14]. It is triangle-free by construction: if $u \prec w \prec v$, the direct link $u \to v$ is removed. Therefore K_4 cannot appear as a subgraph of the Hasse diagram. The densest bipartite structure is $K_{2,n}$—the *Causal Prism*: a past pole u and future pole v connected through $n \geq 3$ mutually spacelike (disconnected) intermediates [10, 15]. Topological mass = n.

Geometry of Matter (Generation Theorem). Each intermediate carries a discrete phase: $\varphi(w_i) = \text{sgn}(p_{\text{bulk}}[w_i]) \in \{-1, 0, +1\}$. The generation of a prism is

$$g(P) = |\{\text{distinct } \varphi(w_i)\}| \;\in\; \{1, 2, 3\}. \tag{4}$$

Exactly three, because the sign function has three values—a theorem, not a fit. The net phase $\Phi(P) = \sum_i \varphi(w_i)$ distinguishes matter ($\Phi \geq 0$) from antimatter ($\Phi < 0$).

K_5 threat absorption enforces confinement: an isolated prism component would create a forbidden K_5 topological minor [10], so prism constituents cannot be separated—mirroring colour confinement in QCD.

The S_n permutation symmetry of the n intermediates (the "belly" of the prism) is not a hint of gauge structure—it *is* the gauge structure. For $n = 3$ intermediates, S_3 is the Weyl group of $SU(3)$: the colour charge algebra of QCD derived from the bipartite combinatorics of the Hasse diagram.

Geometry of Logic. $D \leq 15 \Rightarrow O(N)$. This bound is not imposed—it is a geometric consequence of $\rho \approx 1$ sprinkling into the Hasse diagram of a causal diamond. The simulation's $O(N)$ efficiency is the computational manifestation of physical locality [6, 16].

MACROSCOPIC EMERGENCE

Gravity as information cost. By Chentsov's theorem [8], the unique metric on the space of causal histories is the Fisher information metric: $g_{\mu\nu} \propto I_{\mu\nu}(\theta)$. Gravity is the statistical resistance of the causal network to information deformation. As Jacobson showed [17], when causal flux is bounded by a horizon, $S \propto A/4G$ and the Einstein equations emerge as a thermodynamic equation of state [18, 19].

Dark energy as Poisson noise. Volume fluctuations $\delta V \sim \sqrt{N}$ [20] yield

$$\Lambda \propto \frac{1}{\sqrt{N}} \propto H^2 \tag{5}$$

Dark energy is the statistical jitter of counting causal events; it is tiny because $N \gg 1$ [21].

IMPLICATIONS

By strictly adhering to the Kuratowski Calculus, major anomalies dissolve as artefacts of the continuum assumption:

1. **Singularity:** Holographic capacity bounds Planck volumes to 1 bit; infinite density is obstructed.
2. **Dark energy:** Poisson noise, not vacuum energy (Eq. 5) [20].

3. **Proton stability:** Prisms are topological knots; decay requires reordering $\sim 10^{60}$ links.
4. **Strong CP:** θ is the average of $\sim 10^{60}$ random link phases; by the CLT, $\theta \to 0$.
5. **Baryogenesis:** Forward-growing DAGs favour $\Phi > 0$ (matter) over $\Phi < 0$ (antimatter); the simulation confirms $\sim$13:1.

CONCLUSION

From two axioms and one principle, the $N = 10^7$ simulation produces: (i) three generations bounded by sign trichotomy $g \leq 3$—a theorem; (ii) mass hierarchy $4.56 < 6.54 < 7.73$ without calibration; (iii) CPT to 2.4%; (iv) $\Omega_d/\Omega_v = 2.71$; (v) $d_S = 2 \to 4$ consistent with CDT. $D \leq 15$ makes every operation $O(N)$: the code proves locality. Open: Yang–Mills from S_n; physical mass units; closing the dark matter gap (2.71 vs observed 5.4). All code and data: `https://github.com/ertwro/fractal_entropic_geometrodynamic`.

I dedicate this work to my wife and daughter. I acknowledge the computational endurance of a ThinkPad T480 running Arch Linux, and thank the AI architectures Claude, Grok, and Gemini for serving as adversarial semantic mirrors throughout the synthesis of this framework.

[1] A. Einstein, Die Feldgleichungen der Gravitation, Sitzungsberichte der Preussischen Akademie der Wissenschaften zu Berlin , 844 (1915).
[2] R. M. Wald, *General Relativity* (University of Chicago Press, 1984).
[3] J. Ambjørn, J. Jurkiewicz, and R. Loll, Non-perturbative Lorentzian path integral for gravity, Physical Review Letters **85**, 924 (2000).
[4] R. Loll, Quantum gravity from causal dynamical triangulations: A review, Classical and Quantum Gravity **37**, 013002 (2020).
[5] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, Spacetime as a causal set, Physical Review Letters **59**, 521 (1987).
[6] L. Bombelli, J. Henson, and R. D. Sorkin, Discreteness without symmetry breaking, Modern Physics Letters A **24**, 2579 (2009).
[7] D. Petz, Monotone metrics on matrix spaces, Linear Algebra and its Applications **244**, 81 (1996).
[8] N. N. Chentsov, *Statistical Decision Rules and Optimal Inference*, Translations of Mathematical Monographs, Vol. 53 (American Mathematical Society, 1982) translated from the 1972 Russian original. Proves uniqueness of the Fisher metric under Markov morphisms.
[9] D. M. T. Benincasa and F. Dowker, Scalar curvature of a causal set, Physical Review Letters **104**, 181301 (2010).
[10] K. Kuratowski, Sur le problème des courbes gauches en topologie, Fundamenta Mathematicae **15**, 271 (1930).
[11] A. Eichhorn and S. Mizera, Spectral dimension in causal set quantum gravity, Classical and Quantum Gravity **31**, 125007 (2014).
[12] J. Ambjørn, J. Jurkiewicz, and R. Loll, Spectral dimension of the universe, Physical Review Letters **95**, 171301 (2005).
[13] S. Carlip, Dimension and dimensional reduction in quantum gravity, Classical and Quantum Gravity **34**, 193001 (2017).
[14] S. Surya, The causal set approach to quantum gravity, Living Reviews in Relativity **22**, 5 (2019).
[15] R. Diestel, *Graph Theory*, 5th ed., Graduate Texts in Mathematics, Vol. 173 (Springer, 2017).
[16] M. H. Stone, The theory of representations for Boolean algebras, Transactions of the American Mathematical Society **40**, 37 (1936).
[17] T. Jacobson, Thermodynamics of spacetime: The Einstein equation of state, Physical Review Letters **75**, 1260 (1995).
[18] D. Dou and R. D. Sorkin, Black hole entropy as causal links, Foundations of Physics **33**, 279 (2003).
[19] J. D. Bekenstein, Black holes and entropy, Physical Review D **7**, 2333 (1973).
[20] M. Ahmed, S. Dodelson, P. B. Greene, and R. D. Sorkin, Everpresent Λ, Physical Review D **69**, 103523 (2004).
[21] S. Weinberg, The cosmological constant problem, Reviews of Modern Physics **61**, 1 (1989).